Overview

This study will test whether existing polymerase chain reaction (PCR) assays in Australian laboratories can detect lumpy skin disease (LSD) virus in insects. It will also investigate what conditions affect detection.

Currently, LSD detection during an outbreak relies on collecting samples from animals. Mustering cattle or buffalo can be difficult and costly, especially in remote areas. If insect testing is successful, it could offer a new way to understand the potential spread of LSD during a response. This project will also inform how insect testing could be used for other animal diseases.

Project Lead: Department of Agriculture, Fisheries and Forestry; Australian Centre for Disease Preparedness; Cattle Australia

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Project Status: Active

Project date: June 2025 to December 2026

Objective alignment:

- 1 — Improve Australia's preparedness and ability to respond to emergency animal diseases
- 2 — Improve Australia's surveillance and diagnostic capacity for animal pests and diseases

Activity alignment:

- 1.7. Implement activities identified in the National LSD Action Plan
- 2.2. Develop and implement novel technologies, such as POC animal testing and genomics, to address gaps in diagnostic capacity

Project updates

May 2026

Mosquitoes have been carefully bred, raised and prepared in the Australian Centre for Disease Preparedness (ACDP) insectary for testing. They are now ready to support work to understand the lowest level of LSD virus that can be reliably detected through pooled mosquito testing.

February 2026

Work has commenced at the Australian Centre for Disease Preparedness (ACDP) to refine how mosquito samples are grouped, prepared and tested using existing PCR assays. Studies are also underway to identify whether anything in the samples could interfere with PCR testing.

Laboratory-reared mosquitoes will be used to determine the smallest amount of virus that can be reliably detected in pooled mosquito samples. This work will take place in the ACDP insectary and is planned for April 2026.

See more

Tags: Active, Objective 1, Objective 2